

第一周作业参考解答

p.23, 习题 10.3, 1(2), (4), (5), (6), (7), (8), (9), (11), 2(2), (4), (5), 3(2), (3), (5), 5(1), (2), (5), 6.

1. 解: (2) 令 $x = \rho \cos \theta, y = \rho \sin \theta$, $\theta \in [0, 2\pi]$, 则当 $(x, y) \rightarrow (0, 0)$ 时, 有 $\rho \rightarrow 0$. 由于

$$0 \leq \left| \frac{xy}{\sqrt{x^2 + y^2}} \right| = |\rho \sin \theta \cos \theta| < \rho \rightarrow 0 \quad (\rho \rightarrow 0),$$

故 $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}} = 0$.

(4) 动点沿着直线 $y = x$ 趋近坐标原点时, $\lim_{(x,y) \rightarrow (0,0)} \frac{x+y}{|x|+|y|} = \lim_{x \rightarrow 0} \frac{x}{|x|}$ 不存在, 故极限

$\lim_{(x,y) \rightarrow (0,0)} \frac{x+y}{|x|+|y|}$ 不存在。

(5) 令 $y = kx$, 则

$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} = \lim_{x \rightarrow 0} \frac{(1-k^2)x^2}{(1+k^2)x^2} = \frac{1-k^2}{1+k^2}$ 依赖于 k 的值, 故 $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ 不存在。

(6) 令 $x = \rho \cos \theta, y = \rho \sin \theta$, $\theta \in [0, 2\pi]$, 则当 $(x, y) \rightarrow (0, 0)$ 时, 有 $\rho \rightarrow 0$, 且

$$0 \leq \left| \frac{1 - \cos(xy)}{x^2 + y^2} \right| = \left| \frac{1 - \cos(\rho^2 \cos \theta \sin \theta)}{\rho^2} \right| = |O(\rho^2)| \rightarrow 0 \quad (\rho \rightarrow 0),$$

故 $\lim_{(x,y) \rightarrow (0,0)} \frac{1 - \cos(xy)}{x^2 + y^2} = 0$.

解法二、由于 $(x, y) \rightarrow (0, 0)$ 时, 有 $xy \rightarrow 0$, 且 $0 \leq \left| \frac{1 - \cos(xy)}{x^2 + y^2} \right| \leq \frac{1}{2} \left| \frac{1 - \cos(xy)}{xy} \right|$,

又知 $\lim_{xy \rightarrow 0} \frac{1 - \cos(xy)}{xy} = \lim_{xy \rightarrow 0} \frac{xy}{2} = 0$, 因此双侧趋近定理表明 $\lim_{(x,y) \rightarrow (0,0)} \frac{1 - \cos(xy)}{x^2 + y^2} = 0$.

(7) 令 $y = x$, 则 $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 - y^3}{x + y} = 0$; 再取 $y = x^3 - x$, 则

$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 - y^3}{x + y} = \lim_{x \rightarrow 0} [1 - (x^2 - 1)^3] = 2$, 故 $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 - y^3}{x + y}$ 不存在。

(8) 当 $(x, y) \rightarrow (0, 0)$ 时, 有 $xy \rightarrow 0$. 令 $xy = z$, 则

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy - \sin(xy)}{xy - xy \cos(xy)} = \lim_{z \rightarrow 0} \frac{z - \sin z}{z - z \cos z} = \frac{1}{3}.$$

(9) 取对数, 有 $0 \leq \lim_{(x,y) \rightarrow (0,0)} |xy \ln(x^2 + y^2)| \leq \lim_{(x,y) \rightarrow (0,0)} |(x^2 + y^2) \ln(x^2 + y^2)| = 0$,

故 $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2)^{xy} = 1$.

(11) 令 $x = ky^2$, 则 $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 y^4}{(x^2 + y^4)^3} = \lim_{y \rightarrow 0} \frac{k^4}{(k^2 + 1)^3} = \frac{k^4}{(k^2 + 1)^3}$ 依赖于 k 的值, 所以动点

沿着两条不同曲线 $x = k_1 y^2$ 和 $x = k_2 y^2$ 趋于 $(0, 0)$ 时, $\frac{x^4 y^4}{(x^2 + y^4)^3}$ 的极限存在但不相等, 故

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 y^4}{(x^2 + y^4)^3} \text{ 不存在.}$$

2. 解: (2) 解法一、由于 $x^2 + xy + y^2 \geq 2|xy| + xy \geq |xy|$, 因此

$$0 \leq \left| \frac{x+y}{x^2 + xy + y^2} \right| \leq \left| \frac{x+y}{xy} \right| \leq \frac{1}{|x|} + \frac{1}{|y|} \rightarrow 0 \quad (x \rightarrow \infty, y \rightarrow \infty),$$

$$\text{故 } \lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} \frac{x+y}{x^2 + xy + y^2} = 0.$$

解法二、任意固定充分大的 y , 由于 $\lim_{x \rightarrow \infty} \frac{x+y}{x^2 + xy + y^2} = 0$, 因此 $\lim_{y \rightarrow \infty} \lim_{x \rightarrow \infty} \frac{x+y}{x^2 + xy + y^2} = 0$,

$$\text{故 } \lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} \frac{x+y}{x^2 + xy + y^2} = 0.$$

解法三、令 $x = \rho \cos \theta, y = \rho \sin \theta$, $\theta \in [0, 2\pi]$, 则当 $x \rightarrow \infty, y \rightarrow \infty$ 时, 有 $\rho \rightarrow +\infty$.

$$0 \leq \left| \frac{x+y}{x^2 + xy + y^2} \right| = \left| \frac{\sqrt{2} \cos(\theta + \frac{\pi}{4})}{\rho(1 + \frac{1}{2} \sin 2\theta)} \right| \leq \frac{2\sqrt{2}}{\rho} \rightarrow 0 \quad (\rho \rightarrow +\infty),$$

$$\text{故 } \lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} \frac{x+y}{x^2 + xy + y^2} = 0.$$

$$(4) \lim_{\substack{x \rightarrow \infty \\ y \rightarrow 0}} \left(1 + \frac{1}{x}\right)^{\frac{x^2}{x+y}} = e^{\lim_{\substack{x \rightarrow \infty \\ y \rightarrow 0}} \frac{x}{x+y}} = e.$$

$$(5) \text{ 对 } \forall (x, y) \neq (0, 0), 0 < \frac{|xy|}{x^2 + y^2} \leq \frac{1}{2}, \text{ 而 } \lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} \left(\frac{1}{2}\right)^{x^2} = 0, \text{ 因此 } \lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} \left(\frac{|xy|}{x^2 + y^2}\right)^{x^2} = 0.$$

3. 解: (2) 由于 $\lim_{x \rightarrow +\infty} \lim_{y \rightarrow 0^+} \frac{x^y}{1+x^y} = \frac{1}{2} \neq 1 = \lim_{y \rightarrow 0^+} \lim_{x \rightarrow +\infty} \frac{x^y}{1+x^y}$, 故重极限 $\lim_{\substack{x \rightarrow +\infty \\ y \rightarrow 0^+}} \frac{x^y}{1+x^y}$ 不存在。

(3) 因为

$$0 \leq \left| (x+y) \sin \frac{1}{x} \sin \frac{1}{y} \right| \leq |x| + |y| \rightarrow 0, (x, y) \rightarrow (0, 0),$$

因此重极限 $\lim_{(x,y) \rightarrow (0,0)} (x+y) \sin \frac{1}{x} \sin \frac{1}{y} = 0$. 但两个累次极限 $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} (x+y) \sin \frac{1}{x} \sin \frac{1}{y}$ 和

$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} (x+y) \sin \frac{1}{x} \sin \frac{1}{y}$ 都不存在。

(5) 取 $y = x$, 则 $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 y^2 + (x-y)^2} = \lim_{x \rightarrow 0} \frac{x^4}{x^4} = 1$; 取 $y = kx, k \neq 1$, 有

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 y^2 + (x-y)^2} = \lim_{x \rightarrow 0} \frac{k^2 x^2}{k^2 x^2 + (1-k)^2} = 0,$$

故重极限 $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 y^2 + (x-y)^2}$ 不存在。但两个累次极限都存在,

$$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} \frac{x^2 y^2}{x^2 y^2 + (x-y)^2} = 0, \quad \lim_{y \rightarrow 0} \lim_{x \rightarrow 0} \frac{x^2 y^2}{x^2 y^2 + (x-y)^2} = 0.$$

5. 证明: (1) 令 $y = x$, 则 $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^3 + y^3} = 0$; 取 $y = x(x-1)$, 则

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^3 + y^3} = \lim_{x \rightarrow 0} \frac{x^4 (x-1)^2}{x^3 + x^3 (x-1)^3} = \lim_{(x,y) \rightarrow (0,0)} \frac{(x-1)^2}{x^2 - 3x + 3} = \frac{1}{3} \neq 0,$$

故重极限 $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^3 + y^3}$ 不存在。

(2) 令 $y = x$, 则 $\lim_{(x,y) \rightarrow (0,0)} \left(\frac{xy}{x^2 + y^2} + y \sin \frac{1}{x} \right) = \frac{1}{2}$; 再取 $y = x^2$, 则

$\lim_{(x,y) \rightarrow (0,0)} \left(\frac{xy}{x^2 + y^2} + y \sin \frac{1}{x} \right) = 0$, 故 $\lim_{(x,y) \rightarrow (0,0)} \left(\frac{xy}{x^2 + y^2} + y \sin \frac{1}{x} \right)$ 不存在.

(5) 因为

$$0 \leq \left| \frac{x+y}{x^2 + y^2 + 1} \right| \leq \frac{|x| + |y|}{\frac{1}{2}(|x| + |y|)^2 + 1} \rightarrow 0 \quad (x \rightarrow \infty, y \rightarrow \infty),$$

因此 $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} \left| \frac{x+y}{x^2 + y^2 + 1} \right| = 0$, 从而 $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} \frac{x+y}{x^2 + y^2 + 1} = 0$.

6. 不正确。实际上, 任意固定 $\theta \neq 0, \pi$, 所给出的解答相当于动点沿着任意一条经过坐标原点的直线趋近坐标原点时, 函数极限都存在且等于零, 但保证不了动点以任意方式趋近坐

标原点函数的极限都存在且相等. 例如, 取曲线 $y = kx^2$, 则 $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2} = \frac{k}{1+k^2}$ 依赖

于 k 的值, 所以 $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$ 不存在。